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INVIVO STUDY OF EFFECT OF NIFEDIPINE, KETOTIFEN FUMERATE AND POTASSIUM NITRATE ON PLASMA CONCENTRATION OF DILTIAZEM IN RABBIT

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ABSTRACT

An invivo study has been carried out to evaluate the influence of nifedipine, ketotifen fumerate and potassium nitrate on the plasma concentration of diltiazem hydrochloride in rabbit by spectrophotometric method. It has been found that concurrent administration of nifedipine and ketotifen fumerate increase the plasma concentration of diltiazem hydrochloride but potassium nitrate has hardly any effect on the plasma concentration of diltiazem hydrochloride.

INTRODUCTION

Both diltiazem and nifedipine are calcium channel blocker which have gained increasing acceptance in the treatment of angina pectoris, hypertension and cardiac arrhythmias (Carlsted and Stanazek 1990, Martin 1987). Ketotifen fumerate is an antihistaminic drug used in the treatment of allergic conditions and prophylactic control of asthma (Martindale, the extrapharmacopiea, 1993). Potassium nitrate is a physiologically important metal salt (Gilman et al., 1991). Plasma concentration is the concentration of drug in the plasma in blood (Cadwallader 1985). From drug-plasma (blood) profile we can have an idea about the optimum or desired therapeutic or pharmacological concentration range such as minimum effective concentration (MED), minimum toxic concentration (MTC) etc. MEC represents the minimum effective concentration at the site of action. With the help of MEC some others parameters such as onset, duration of action, maximum safe concentration, amount observed etc. can be determined. Hussain et al. (1992). Studied a sensitive, specific and reproducible HPLC technique for simultaneous determination in human plasma of diltiazem and six of its primary and secondary metabilities. V. Ascalone (1994) investigated the determination of diltiazem and its main metabolites in human plasma by automated solid phase extraction and HPLC. K.F.P. Yeung (1989) studied the HPLC assay of diltiazem and six of its metabolites to determine the kinetics of diltiazem and its main metabolites in healthy volunteers after each received a single 90 mg oral dose of diltiazem. Chacko Verghese (1983) studied the HPLC analysis of diltiazem and its main metabolites in plasma.

The aim of the present study is to evaluate the interaction of nifedipine and ketotifen fumerate with diltiazem through studying the effect of nifedipine and ketotifen fumerate on the plasma concentration of diltiazem.

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EXPERIMENTAL

Apparatus:

A uv-vis recording spectrophotometer (UV-160, Shimadzu, Japan) was used for the measurement of absorbance of the serum samples. A centrifuge machine was used to centrifuge the plasma samples.

Chemicals:

Diltiazem hydrochloride and ketotifen fumerate were obtained from Beximco Pharmaceuticals Ltd. (Bangladesh). Nifedipine was kindly supplied by ACME Laboratories Ltd. (Bangladesh). Potassium nitrate was of analytical grade and heparin was purchased from Leo pharmaceutical products, Denmark.

Test animals:

The adult healthy rabbits weighing about 2 + 0.1 Kg were used as the experimental animals. The animals were obtained from the International Center for Diarrheal Disease Research, Bangladehs (ICDDR, B).

Method

The UV spectrophotometric method was used to measure the plasma concentration. Ten adult healthy rabbits of 2 + 0.1 Kg body weight were used. Rabbits were kept rest for 7 days with normal diets. Nine rabbits were divided into three groups each having three (marked as I, II, III) and one as control. 2 mg of diltiazem alone and its 1:1 molar mixtures with nifedipine, ketotifen and potassium nitrate were administered by orogastric tube individually. They were overnight fasted before drug administration. Venous blood samples were collected from the ear vein into heparinized centrifuge tubes before drug administration and at 0.5, 1.0 2.0, 3.0, 4.0, 6.0 hour after drug administration. All blood samples were protected from light, immediately centrifuged at 3000 rev/min for 10 minutes and the plasma samples were separated into vials and kept into deep freeze upto taking absorbance.

RESULTS AND DISCUSSION

Plasma concentration of diltiazem HCl was determined by UV spectrophotometer after oral single administration of diltiazem HCl (2 mg_ alone and with nifedipine (2 mg each), ketotifen fumerate (2 mg each) and KNO_3 (2 mg each) in rabbit by using a standard curve.

It was found that the peak plasma concentration of diltiazem was 78.50 ng/ml which was obtained after two hour of oral administration of diltiazem alone (Fig. 1). This peak time is within the normal range. Concurrent administration of diltiazem HCl and nifedipine made a significant change in plasma concentration of diltiazem HCl. In this case the peak plasma concentration was 107 ng/ml which is significantly greater than that of diltiazem HCl when administered alone (Fig. 2). This may be due to higher affinity of nifedipine for protein (the plasma protein binding of nifedipine is 92-98%). So competitive inhibition to the binding of plasma protein by nifedipine increase the plasma concentration of diltiazem HCl.

Concurrent administration of diltiazem and ketotifen fumerate also made a significant change in the plasma concentration of diltiazem HCl, the peak plasma concentration being 94.5 ng/ml (Fig. 3). This may also be due to higher affinity of ketotifen to the plasma protein.

Oral concurrent administration of diltiazem and KNO_3 did not make a noticeable change in plasma concentration of diltiazem HCl. The peak plasma level was 75.5 ng/ml (Fig. 4). This indicates that KNO_3 does not make any intervention in the plasma protein binding of diltiazem and thus cause hardly any change in plasma concentration of diltiazem. These results are comparable to that of Bilkis and Coworker (1996).

Such interactions of drugs that affect the binding of plasma protein and subsequently that change the plasma concentration of drugs are very vital to be given priority in the combined drug therapy. Since drug displaced from plasma protein will redistribute into its full potential volume of distribution, the concentration of free drug in plasma and tissue after redistribution may change the pharmacokinetic and pharmacodynamic properties of the drug and thereby may affect its pharmacological and toxic effects. Thus great care must be taken during combination therapy with these drugs.

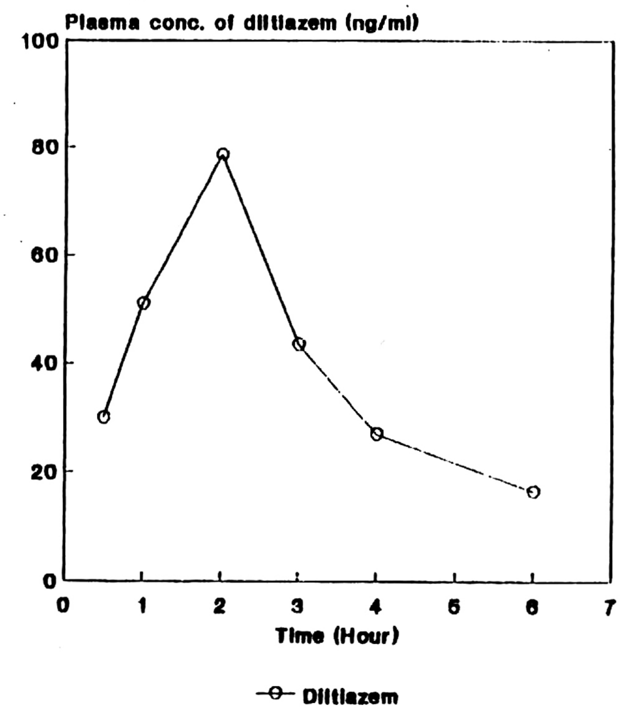

Fig. 1: Plasma concentration of diltiazem when administered alone.

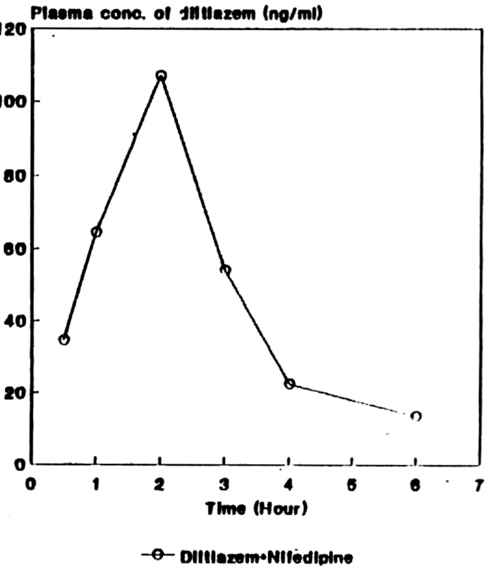

Fig. 2: Plasma concentration of diltiazem after coadministration with nifedipine.

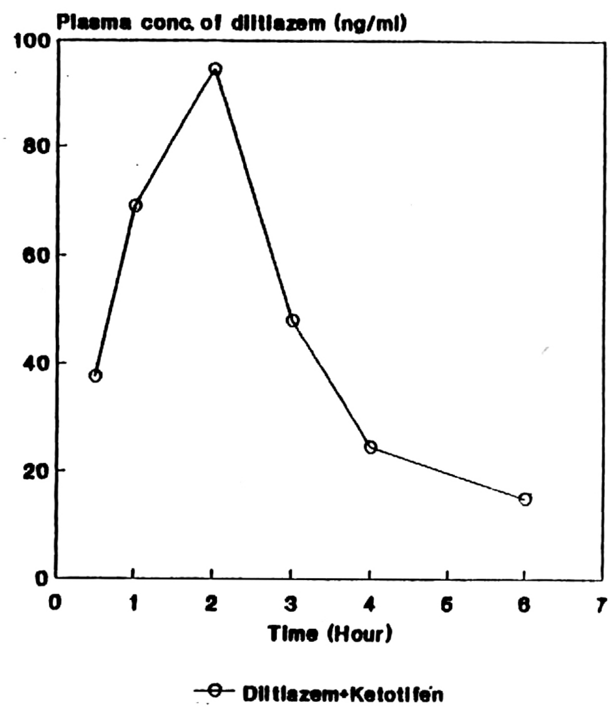

Fig. 3: Plasma concentration of diltiazem after coadministration with ketotifen fumerate.

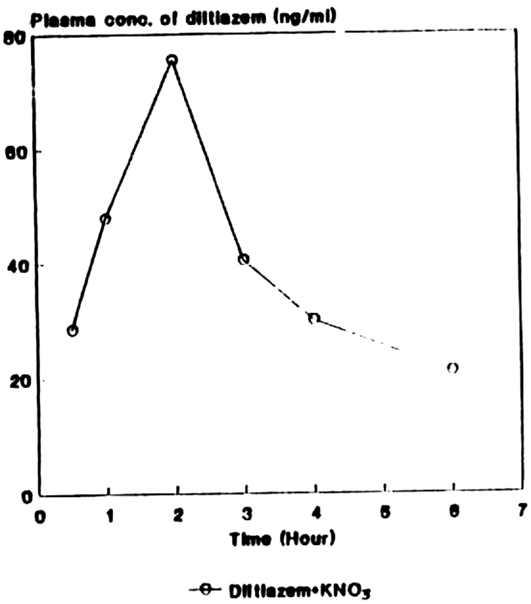

Fig. 4: Plasma concentration of diltiazem after coadministration with potassium nitrate.

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